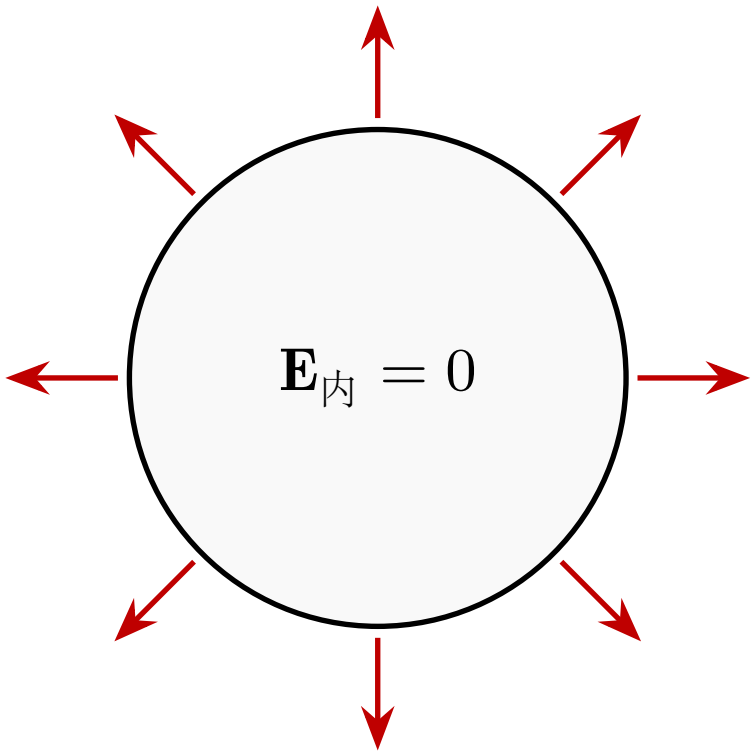
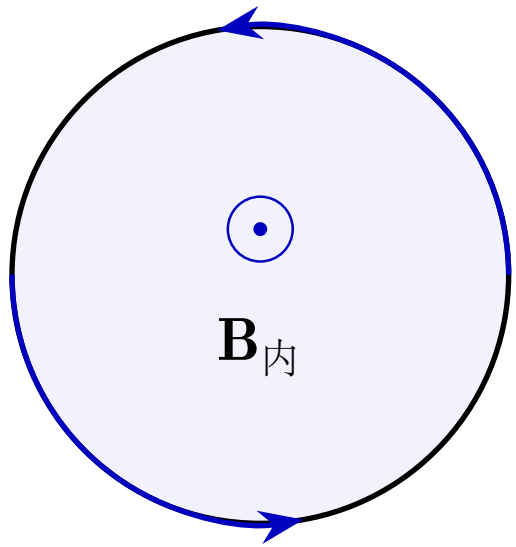
从 $+z$ 看，图示取 $\sigma > 0,\ \omega > 0$

电场



$$\mathbf{E}_{外} = \frac{\sigma}{\varepsilon_0}\mathbf{e}_r$$

面电流与磁场



$\mathbf{B}_{外} = 0$

$$\mathbf{B}_{内} = \mu_0 \sigma \omega r\, \mathbf{e}_z$$